SpcTopics Assignment 1: Glucose Control Part 1

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1 Introduction

A toy model of glycemic control is a simplified mathematical representation of how the human body regulates blood glucose levels. By neglecting some of the complexities of biological systems, the model captures the fundamental dynamics of glucose regulation [1]. The model introduced in Lecture 4 is described by the following equations.

$$u = \alpha\bar{v} + \beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t')\, dt'$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t).$$

Here, v represents the deviation of glucose concentration from its baseline value, $\bar{v}$ represents the baseline glucose concentration, and u is a control variable that influences glucose regulation. The parameter γ represents the rate at which glucose is consumed by basic bodily functions, while $F(t)$ represents external influences on glucose concentration, such as food intake or physical exertion. The parameters α, β, λ, and δ determine the strength and dynamics of the control response.

The model incorporates a history-dependent control mechanism through the integral term in the first equation. This term calculates an exponentially weighted average of past glucose deviations, giving greater weight to more recent values. The parameter λ determines how quickly the influence of past glucose deviations decays over time. The resulting control variable u affects glucose dynamics through the second equation, where the term $-\delta u(v + \bar{v})$ represents the control-dependent contribution to glucose removal.

2 Derive an ODE with 2 DOF

Starting with the toy control system, the integral must be differentiated for u, then replace the integral with the same equation.

Let:

$$I(t) = \int_{-\infty}^{t} \lambda e^{-\lambda(t-s)} v(s) ds$$

Then:

$$u = \alpha v + \beta I$$

Now to differentiate the integral, the upper limit moves with t and the exponential inside also depends on t. Since $e^0 = 1$, and the remaining integral is $-\lambda I(t)$, let:

$$\dot{I} = \lambda v - \lambda I$$

Then differentiate u and substitute out I, simplify, and replace $\dot{v}$ with its given expression:

$$\dot{u} = -\lambda u + \lambda(\alpha + \beta)v + \alpha F(u, v)$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F$$

3 Reducing the Number of Parameters

Recall the reduced equations from the previous section, because to reduce the number of parameters the equations will be nondimensionalized, using the scales below.

$$v = \bar{v}v' -> \frac{v}{\bar{v}} = v'$$
$$u = \tilde{u}u' -> \frac{u}{\tilde{u}} = u'$$
$$t = \tilde{t}t' -> \frac{t}{\tilde{t}} = t'$$

Then scale the derivatives using the chain rule in the system of equations. First, with equation $\dot{v}$, plug in the scales, and simplify.

$$\frac{1}{\tilde{t}}\bar{v}\frac{dv'}{dt'} = -\gamma(\bar{v}v') - \delta(\tilde{u}u')(\bar{v}v' + \bar{v}) + F$$

$$\frac{dv'}{dt'} = -(\gamma\tilde{t})v' - (\delta\tilde{t}\tilde{u})u'(v' + 1) + \frac{\tilde{t}}{\bar{v}}F$$

Note: Let $G = \frac{\tilde{t}}{\bar{v}}F$
Rewrite $\gamma\tilde{t} = 1$ as $\tilde{t} = \frac{1}{\gamma}$, and $\delta\tilde{t}\tilde{u} = 1$ as $\tilde{u} = \frac{\gamma}{\delta}$.
Therefore:

$$\frac{dv'}{dt'} = -v' - u'(v' + 1) + G$$

As for the next equation, $\dot{u}$, the scales are plugged in and simplified. Steps not shown, but similar to the ones for $\dot{v}$.

$$\frac{du'}{dt'} = -\frac{\lambda}{\gamma}u' + [(\frac{\lambda\gamma}{\delta})(\alpha + \beta]v' + (\frac{\alpha}{\delta})f'(u', v')$$

Let $a = \frac{\lambda}{\gamma}, b = \frac{\lambda\bar{v}}{\delta}(\alpha + \beta), c = \frac{\alpha}{\delta}$
Therefore:

$$\frac{du'}{dt'} = -au' + bv' + cf'(u', v')$$

4 Equilibrium Solutions

First, drop the primes since everything is dimensionless now, and set the two equations to zero.

$$-au + bv = 0$$
$$u = \frac{b}{a}v$$

$$-v - u(v + 1) + G = 0$$
$$-v - \frac{b}{a}v(v + 1) + G = 0$$

$$\boxed{v^* = -\frac{1}{2}\left(1 + \frac{a}{b}\right) \pm \frac{1}{2}\sqrt{\left(1 + \frac{a}{b}\right)^2 + \frac{4aG}{b}}, \qquad u^* = \frac{b}{a}v^*.}$$

The quadratic formula gives two branches. The real equilibria require the expression under the square root to be nonnegative. The branches meet when it is zero:

$$\boxed{G_{SN} = -\left(1 + \frac{a}{b}\right)^2 \frac{b}{4a}.}$$

This equilibrium branches meet at a nondegenerate fold, where the Jacobian deteminant vanishes. Provided the Jacobian trace is nonzero, this point is a saddle-node bifurcation: two equilibria merge and disappear as G decreases through G_{SN}.

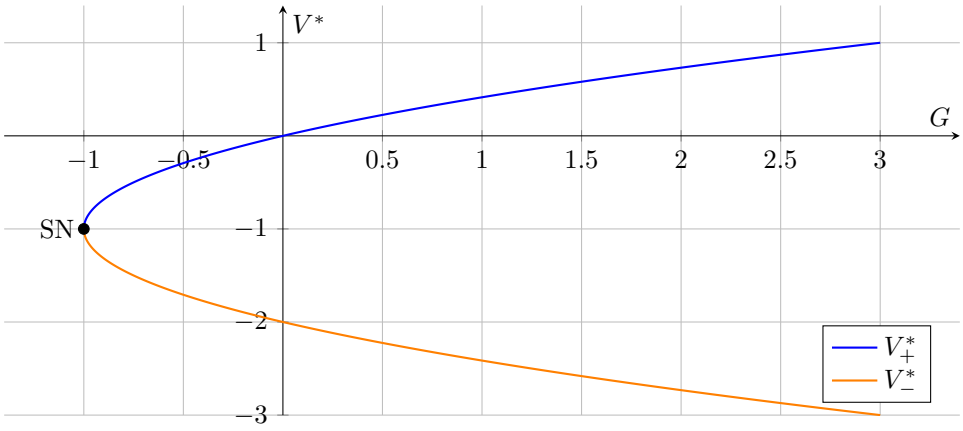


Figure 1: Equilibrium branches for $a = b = 1$. The branches meet at $G_{\text{SN}} = -1$. There are no equilibria for $G < -1$.

5 Exploring Initial Conditions

A controllable model is a system that the available control input can be used to drive the system from its current state to a desired state, subject to the model's constraints. To explore what parameters and initial conditions make the model controllable, an experimental python file was created. The file calls in the original python file with the corresponding rhs, jacobian, and runge-kutta functions. The parameters for the system and time step are set, and a simulation function was created.

Three experiments were conducted. Experiment 1 varied the initial value v_0 while keeping u_0 fixed to investigate the system's long-term behavior. Experiment 2 varied p_3, which represents the input parameter G, to examine how changes in the external input affect the existence of equilibria and the resulting dynamics. Experiment 3 varied p_0, the coefficient governing the decay of u, to investigate how the strength of decay affects the system's stability and convergence.

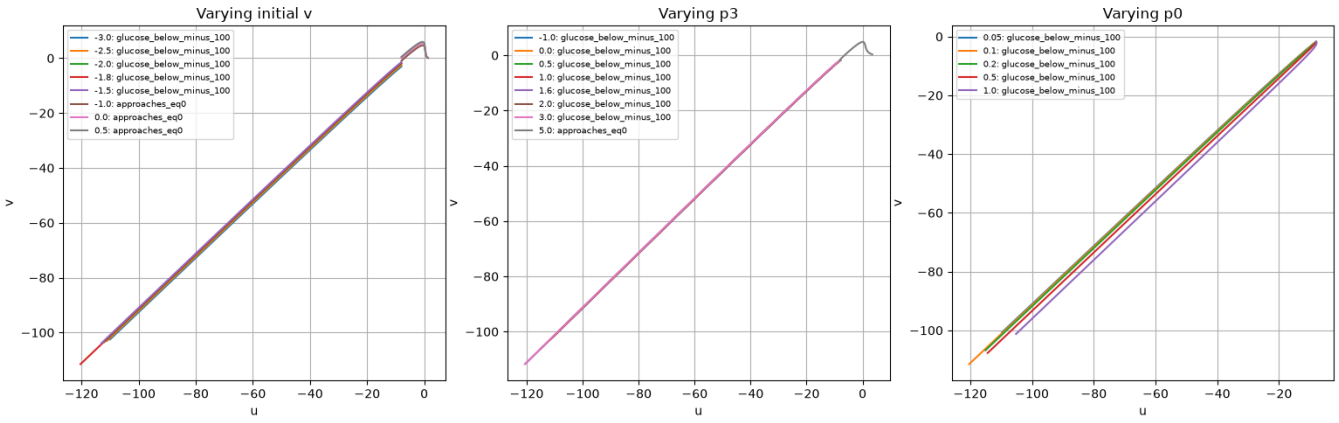


Figure 2: The trajectories plotted in the phase plane.

Figure 2 shows the trajectories in the phase plane. There are two kinds of behavior. In the first, the trajectory moves down and to the left along nearly the same straight line until it reaches the cutoff $v = -100$. In this case the glucose keeps dropping, so the model is not controlled. This happens for $v_0 \leq -1.5$, for $p_3 \leq 3$, and for every value of p_0 tested. In the second kind, the trajectory rises quickly, overshoots to about $v \approx 5$ near $u \approx 0$, and then settles at the stable equilibrium. This happens for $v_0 \geq -1.0$ and for $p_3 = 5$.

Figure 3 shows the final value of v for each experiment. The points near 0 are the controlled runs, where v settles at the equilibrium value (about 0.13, or 0.35 when $p_3 = 5$). The points between -100 and -112 are the runs that are not controlled. The differences between these points come from the last time step passing the cutoff by different amounts, so they do not show a real trend. Therefore, the useful information is which group each point is in. With $p = [0.1, 1, 1, 1.6]$ and $u_0 = -8$, the model is controlled for $v_0 \geq -1.0$ and not controlled for $v_0 \leq -1.5$. With the initial condition $(-8, -1.8)$, the model is not controlled for $p_3 \leq 3$ and is controlled for $p_3 = 5$, so the cutoff value of

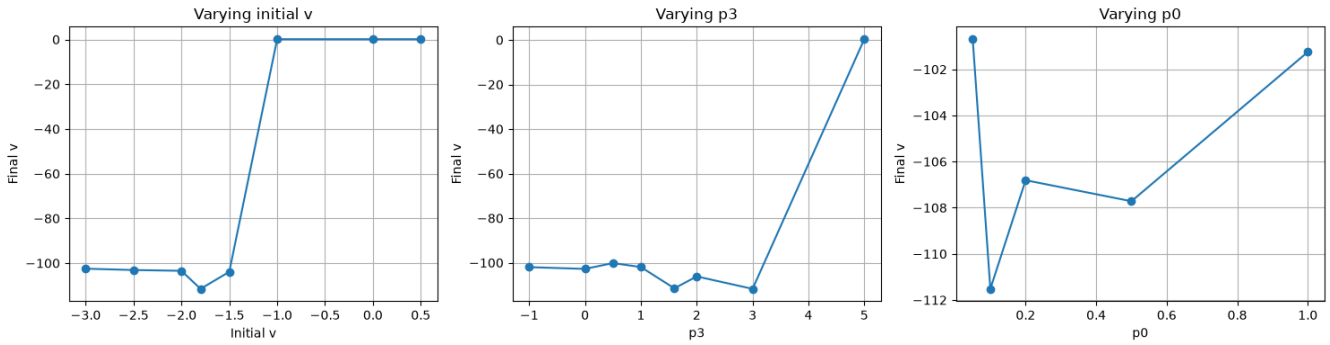


Figure 3: The final v plotted against the parameter or initial condition.

p_3 is between 3 and 5. For $0.05 \leq p_0 \leq 1.0$, the model is not controlled for this initial condition. These results show that there is a sharp threshold: a low enough initial glucose or a low enough input p_3 leads to the glucose dropping instead of recovering.

References

[1] Gabriela Urbina, Daniel N. Riahi, and Dambaru Bhatta. "Mathematical modeling of nonlinear blood glucose-insulin dynamics with beta cells effect". In: *Applications and Applied Mathematics: An International Journal* 15.1 (2020), pp. 171–191.